

Fractional-perimeter counterexamples to the negative- γ BV liminf questions

FA/Banach run `fa_banach_001`, lane 1

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Source metadata and status

The source is H. Brezis, A. Seeger, J. Van Schaftingen, and P.-L. Yung, *Sobolev spaces revisited*, arXiv:2202.01410v1 (2022). The open questions occur on page 20, in Section 6. The local source is `source_paper.pdf`.

Claimed status: counterexample, likely valid. For every dimension $n \geq 1$ and every $-1 < \gamma < 0$, both negative- γ questions quoted below have negative answers. The argument does not address $\gamma \leq -1$ or $\gamma > 0$.

The source questions

For $\gamma \in \mathbb{R}$, put

$$d\nu_\gamma(x, h) = |h|^{\gamma-n} dx dh, \quad \mathcal{E}_{\lambda, 1+\gamma}[u] = \left\{ (x, h) : \frac{|u(x+h) - u(x)|}{|h|^{1+\gamma}} > \lambda \right\}.$$

The source asks, for $\gamma < 0$:

If $u \in L^1_{\text{loc}}$ and

$$\liminf_{\lambda \rightarrow 0^+} \lambda \nu_\gamma(\mathcal{E}_{\lambda, 1+\gamma}[u]) < \infty,$$

must $u \in \dot{BV}$? If $u \in \dot{BV}$, must $\|u\|_{\dot{BV}}$ be bounded by a multiple, depending only on n and γ , of this liminf?

A number of interesting questions remain regarding this circle of ideas. For instance, for $\gamma < 0$, if $u \in L^1_{\text{loc}}$ and the liminf on the right hand side of (19) is finite, must it be true that $u \in \dot{BV}$? If $u \in \dot{BV}$, must $\|u\|_{\dot{BV}}$ be bounded by a multiple (depending only on n and γ) of the liminf on the right hand side of (19)? Some similar questions remain open for $\gamma > 0$ and for $\dot{W}^{1,p}$ in place of $\dot{BV}$. For $\gamma \in \mathbb{R} \setminus \{0\}$, can one understand the failure of (16) for $u \in \dot{BV}$ using the concept of Γ -convergence? A more detailed description of these questions, including discussions of partial results, can be found in [6, Section 7]. Also it is conceivable that one may be able to recover the Sobolev inequality, and its Lorentz refinement, namely

$$\|u\|_{L^{p^*, p}} \lesssim \|\nabla u\|_{L^p}, \quad 1 \leq p < n, \quad \frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$$

out of Theorem 4.

Proof intuition

Write $s = -\gamma$. On the interval $-1 < \gamma < 0$ one has $0 < s < 1$. For a characteristic function $u = \mathbf{1}_A$, the superlevel indicator is bounded by $|u(x+h) - u(x)|$. Consequently the questioned quantity is at most λ times the fractional $W^{s,1}$ seminorm of u , and therefore tends to zero whenever A has finite fractional s -perimeter.

This observation supplies both counterexamples. A cube has positive finite classical perimeter and finite fractional perimeter, so it destroys the proposed lower bound for the BV norm. For the membership question, take a bounded union of infinitely many disjoint intervals whose lengths decay so fast that the sum of their fractional perimeters is finite. It has infinitely many classical boundary points and hence infinite variation. Taking its Cartesian product with a cube gives the same construction in every dimension.

Counterexample theorem

Theorem 1. *Let $n \geq 1$ and $-1 < \gamma < 0$. Then:*

1. *There is a compactly supported $u \in L^1(\mathbb{R}^n) \setminus \dot{BV}(\mathbb{R}^n)$ for which*

$$\lim_{\lambda \rightarrow 0^+} \lambda \nu_\gamma(\mathcal{E}_{\lambda,1+\gamma}[u]) = 0.$$

In particular, finiteness of the liminf does not imply membership in $\dot{BV}$.

2. *There is a nonzero compactly supported $v \in \dot{BV}(\mathbb{R}^n)$ with*

$$\lim_{\lambda \rightarrow 0^+} \lambda \nu_\gamma(\mathcal{E}_{\lambda,1+\gamma}[v]) = 0.$$

Thus no finite constant $C(n, \gamma)$ can bound $\|v\|_{\dot{BV}}$ by the questioned liminf for all $v \in \dot{BV}$.

Both u and v may be chosen to be characteristic functions of bounded measurable sets.

Proof

Set $s = -\gamma \in (0, 1)$ and use the homogeneous fractional seminorm

$$[f]_{\dot{W}^{s,1}(\mathbb{R}^n)} := \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \frac{|f(x+h) - f(x)|}{|h|^{n+s}} dx dh.$$

Lemma 1. *If $A \subset \mathbb{R}^n$ is measurable and $\mathbf{1}_A \in \dot{W}^{s,1}(\mathbb{R}^n)$, then*

$$0 \leq \lambda \nu_{-s}(\mathcal{E}_{\lambda,1-s}[\mathbf{1}_A]) \leq \lambda [\mathbf{1}_A]_{\dot{W}^{s,1}(\mathbb{R}^n)}$$

for every $\lambda > 0$. In particular, the expression tends to zero as $\lambda \rightarrow 0^+$.

Proof. The difference of two values of $\mathbf{1}_A$ is either zero or one. Hence

$$\mathbf{1}_{\mathcal{E}_{\lambda,1-s}[\mathbf{1}_A]}(x, h) \leq |\mathbf{1}_A(x+h) - \mathbf{1}_A(x)|.$$

Since $d\nu_{-s} = |h|^{-n-s} dx dh$, integration and multiplication by λ give the assertion. $\square$

We first construct the non-BV set. In one dimension let

$$I_k = \left(2^{-k}, 2^{-k} + 2^{-2k-2}\right), \quad E = \bigcup_{k=1}^{\infty} I_k.$$

The intervals are pairwise disjoint. If I is an interval of length ℓ , a direct integration gives

$$[\mathbf{1}_I]_{\dot{W}^{s,1}(\mathbb{R})} = \frac{4}{s(1-s)} \ell^{1-s}. \quad (1)$$

Because $\mathbf{1}_E = \sum_k \mathbf{1}_{I_k}$ pointwise and the intervals are disjoint, the triangle inequality and Tonelli's theorem give

$$[\mathbf{1}_E]_{\dot{W}^{s,1}(\mathbb{R})} \leq \sum_{k=1}^{\infty} [\mathbf{1}_{I_k}]_{\dot{W}^{s,1}(\mathbb{R})} = \frac{4}{s(1-s)} \sum_{k=1}^{\infty} 2^{-(2k+2)(1-s)} < \infty.$$

On the other hand, $\mathbf{1}_E$ has infinite variation. Indeed, for every m one can choose pairwise disjoint neighborhoods of $I_1, \dots, I_m$ and a function $\phi_m \in C_c^1(\mathbb{R})$, $|\phi_m| \leq 1$, supported in their union, such that ϕ_m equals -1 at the left endpoint and 1 at the right endpoint of each of these intervals. Therefore

$$\int_E \phi'_m(t) dt = 2m.$$

The variational definition of BV now shows that $\mathbf{1}_E \notin BV(\mathbb{R})$.

For $n > 1$, let $Q = (0, 1)^{n-1}$ and $A = E \times Q$; for $n = 1$, simply set $A = E$. We verify that A still has finite fractional perimeter. With $h = (t, z) \in \mathbb{R} \times \mathbb{R}^{n-1}$,

$$|A \triangle (A - h)| \leq |Q| |E \triangle (E - t)| + |E| |Q \triangle (Q - z)|.$$

Integrating this estimate against $|h|^{-n-s}$ and first integrating in z in the first term and in t in the second gives

$$[\mathbf{1}_A]_{\dot{W}^{s,1}(\mathbb{R}^n)} \leq C_{n,s} \left(|Q| [\mathbf{1}_E]_{\dot{W}^{s,1}(\mathbb{R})} + |E| [\mathbf{1}_Q]_{\dot{W}^{s,1}(\mathbb{R}^{n-1})} \right) < \infty.$$

For completeness, the last seminorm is finite because $|Q \triangle (Q - z)| \leq C_n \min\{|z|, 1\}$; its integral against $|z|^{-(n-1)-s}$ is integrable both at zero (since $s < 1$) and at infinity (since $s > 0$).

The set A has infinite classical perimeter. To see this directly, fix $0 \leq \psi \in C_c^1(Q)$ with $\int_Q \psi > 0$ and $\|\psi\|_{\infty} \leq 1$. The vector fields

$$\Phi_m(x_1, x') = (\phi_m(x_1)\psi(x'), 0, \dots, 0)$$

have supremum norm at most one, while

$$\int_A \operatorname{div} \Phi_m = 2m \int_Q \psi \longrightarrow \infty.$$

Thus $u = \mathbf{1}_A$ lies in $L^1(\mathbb{R}^n) \setminus BV(\mathbb{R}^n)$. Lemma 1 proves part (1).

For part (2), take $B = (0, 1)^n$ and $v = \mathbf{1}_B$. This is a nonconstant BV function and $\|v\|_{BV} > 0$. Moreover,

$$|B \triangle (B - h)| \leq C_n \min\{|h|, 1\},$$

so exactly the same radial integration shows that $[v]_{\dot{W}^{s,1}(\mathbb{R}^n)} < \infty$. Lemma 1 yields a zero limit, proving part (2). $\square$

Verification notes

The proof uses only nonnegative integrals and exact inequalities. The main checks are:

- $\gamma = -s$ makes the measure kernel exactly $|h|^{-n-s}$;
- $1 + \gamma = 1 - s > 0$ is not needed for the domination in Lemma 1, but $0 < s < 1$ is essential for the existence of nonconstant characteristic functions with finite fractional seminorm;
- the interval series converges for every $s < 1$ and its classical variation is infinite;
- the product estimate proves the claim in every dimension without an appeal to a slicing theorem.

No numerical or computer-assisted step is used.

Scope, novelty check, and review recommendation

This packet completely answers both quoted questions negatively on the whole parameter interval $-1 < \gamma < 0$, for every $n \geq 1$. It does not settle the remaining ranges. In particular, the source notes that the membership/lower bound statement at $\gamma = -1$ is known to behave differently; the present fractional-perimeter construction becomes critical there.

The run indexes were searched for arXiv:2202.01410, the title, the exact liminf questions, “fractional perimeter”, and close BV variants. They contained no prior packet for this result. Bounded web searches on July 21, 2026 used the exact functional notation, the source title and quoted question, and combinations of “fractional perimeter”, “liminf”, “BV”, and “negative gamma”. They found the two source papers but no later paper stating this counterexample. Novelty confidence is moderate: the mechanism is elementary and may be folklore even though no matching written answer was found.

Human review recommendation. Review as a likely valid counterexample. The decisive line is Lemma 1; reviewers should also confirm that the source intended the questions parameter-by-parameter, as its surrounding discussion indicates.

References

- [1] H. Brezis, A. Seeger, J. Van Schaftingen, and P.-L. Yung, *Sobolev spaces revisited*, arXiv:2202.01410, 2022.
- [2] H. Brezis, A. Seeger, J. Van Schaftingen, and P.-L. Yung, *Families of functionals representing Sobolev norms*, arXiv:2109.02930; *Analysis & PDE* **17** (2024), 943–979.